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Application of Xgboost Algorithm for Sales Forecasting Using Walmart Dataset

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Abstract. Forecasts have historically acted as the foundation for supply chain preparation and execution. Forecasts are at the heart of supply chain decisions, and they've never been more relevant than they are now, thanks to rising consumer demands, shorter lead times, and the need to handle finite capital. Advances of technologies and data processing processes have culminated in the production of massive quantities of data on a wide range of topics and at a rapid rate over the last ten years. Any shopping mall in today's digitally linked world wants to anticipate consumer needs in order to prevent seasonal shortages of selling products. Companies and shopping centres are getting better at forecasting retail prices and consumer needs on a daily basis. For reliable revenue forecasting, extensive analysis is being conducted at the enterprise stage. Since a company's profit is directly proportional to its ability to forecast profits accurately, Big Marts are looking for a more reliable prediction algorithm to ensure that the company does not lose money. This study explore how Extreme Gradient Boosting (XGBoost) can be used to forecast sales. Walmart stores sales data that was collected from 45 stores were used to train and test the model. Results obtained from the mean absolute error, R2, mean square error and root mean square error revealed that XGBoost machine learning technique can be effectively used to predict sales and this could guide sales managers and other practitioners in making decisions about how to fix prices of goods.

Keywords: Sales Forecasting, XGBoost Algorithm, Machine Learning, Walmart Dataset

1 Introduction

The importance of sales forecasting cut across all phases of a company's supply chain. Sales forecasting is critical for manufacturing, logistics, and decision-making [1]. Because of shorter lead times, the consumer demands, and the need to manage scarce resources, forecasts have become extremely relevant. Market forecasting that is accurate normally results in lower inventory costs and better customer service. Accurate forecasting is also critical in retail and industrial activities [2]. Forecasting is used in many processes and management decisions that influence a company's profitability [3]. For example, long-term forecasting is used in production preparation to assess a sufficient amount of personnel and as an input for business planning, such as planning for expansion or reduction of production units. When a company wants to incorporate a forecasting framework, it's common to copy ideas, goals, and values from other firms to speed up the process. But, since most forecasting approaches were designed for commercial goods, there's a danger of adopting unreasonable targets and unsuitable error measures when the world changes [4]. Inaccurate demand predictions, for

example, have resulted in suboptimal levels of production workforce, resulting in workload imbalances and rises in the costs of recruiting, firing, and overtime labour activities; in many situations, outsourced commodity procurement and inaccurate forecasts may result in a shortage of production supplies. As a result, fast reactions to sales results along with forecasting error delivery help to reduce market risks while still maintaining the balance between manufacturing materials and inventory. In periods of crises, unexpected events, and volatile market conditions, certain responses become ever more important. According to authors in [5], long-term, exponentially emerging disasters, such as COVID-19 (Coronavirus), perplex policymakers and decision-makers, and profoundly disturb population activities and supply chains; COVID-19's novelty and uncertainty make it difficult to deal with, resulting in resource shortages and supply chain disturbances. Furthermore, lockdowns and restricted goods flow render delivery centers unavailable, disrupting the entire supply chain [6]. Traditionally, forecasting analysis aimed to find the best model for a given collection of results [7].

Sales forecasts can be done in a variety of ways, depending on the situation. Traditional time series approaches, such as the ARIMA model and the Holt-Winters technique, have traditionally been used to forecast sales but recent AI-based methods have piqued interest due to their potential to improve predictive accuracy and model nonlinear trends [8,9,10,11]. Orthodox predictive forecasting techniques, according to [1], extrapolate past patterns and seasonal variations to estimate the future, and as a result, these methods are incapable of predicting environmental macroeconomic developments in the market, which usually have a major impact on production. Firms either manually changed their mathematical forecasts or relied on expert judgmental forecasts to deal with these changes. These tactics, though, are skewed since humans are slow at making certain changes, and the method is time consuming. However, machine learning algorithms have shown their prowess as a forecasting technique whose outputs can be used to make informed decision about sales prediction and forecasting. This work studied the application of Extreme Gradient Boosting (XGBoost) algorithm for sales forecasting. Machine learning algorithms are probability based. They provide the tendency of an event occurring in percentage. This probability is derived from data fed into the system. Machine learning is a subset of Artificial intelligence which is employed in so many industries in developing; self-driven cars, speech recognition, and speech-to-text applications. Sales Forecasting is one of the areas where machine learning is applied in predicting future sales and demand using learnt data. Probability occurs mainly in two ways either 0 or 1 (yes or no). The present and past data will be processed using machine learning putting all factors whether externally or internally into consideration in making the best decisions for the sales process. This is done through Long Short-Term Memory (LSTM); which uses neural network for making predictions with provided sequential data. Also, Regressive Models can be used which include applications in Linear regression, Random Forest Regression and XGBoost. Modelling and comparing models are used by looking into the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) which are used in comparing model performance. This type of probability-based decisions helps sales leaders in understanding how natural factors like rainfall, seasons or man-made factors e.g. Low budget capital, economic recession limited staff and resources affects sales process in business. However, this study explore how XGBoost can be used to forecast sales. Walmart stores sales data that was collected from 45 stores were used to train and test the model.

2 Related Works

Since top officials of organizations are the ones to give the final verdicts if the results of a sales forecast will be embraced or discarded, authors in [12] studied the perception of sales managers to sales forecasting. The views of 400 Business - to - business sales managers were retrieved and analyzed, this were then evenly divided using gender, to find out their forecasting priorities, values, and wishes. furthermore, gender differences were examined and recognize several key female attitudes that differ from those of their male counterparts were noted. the views of women's and men's, habits, and demographics were also retrieved and analyzed. The study revealed that when it comes to their forecasting

roles, women and men have equal perceptions. The discrepancies that do exist, however, emphasize the importance of ongoing preparation and instruction, as well as the advancement of constructive organizational cultures and practices. Given the emphasis of the paper on forecasts and the outdated myth that work functions involving numbers are best fit for men, the survey items in the analysis that classify men and women of the same expertise identify a more even playing field, providing optimism and confidence that sales are progressing favorably in terms of gender equity. Authors in [13] carried out a Two-Level statistical model for big mart sales prediction. A two-level methodology was used to forecast product sales from a specific outlet, which produced better predictive results than any of the common single model predictive learning algorithms. The method was tested using 2013 revenue data from big marts. In order to forecast reliable performance, data discovery, data transformation, and feature engineering became crucial. The results showed that a two-level statistical approach outperformed a single model approach because the former had more detail, which resulted in better prediction.

Furthermore, Artificial Neural Networks and Analytical Hierarchy Process were employed for predicting car sales in [14]. In order to elicit factors impacting car sales in North America, a questionnaire was provided to car sellers. The questionnaire variables were then given weights and fed into the proposed neural network as data. To compare the degree of error and achieve a more suitable final output that is similar to truth, linear and exponential regression methods were chosen. The obtained result shows the neural network's good efficiency as compared to other approaches, and it was discovered that it has a lower Minimum Square Error (MSE) than other techniques. Moreover, a study aimed at forecasting sales of electric vehicles was carried out in [15]. To derive further value from the available observations, a new self-adaptive optimized grey model was used, which uses a complex weighted series. The proposed model's weighted coefficient and adjusted initial condition will change to different samples, increasing the model's applicability. The history value was then reconstructed using Simpson's algorithm to improve forecasting accuracy. The modification was discovered to improve the grey model's adaptability and prediction accuracy, making it ideal for predicting electric vehicles. A similar study was carried out in [16]. The genetic algorithm was used to predict the sales of energy vehicles. The integrated buffer operator greatly improves the adaptability of the grey model to the production and sales data of new energy vehicles in China, and has a much higher estimation accuracy than the classical buffer operator and grey model, according to the forecasting findings. Furthermore, the results of the forecast show that from 2018 to 2020, China's new energy vehicle production and sales will rise at an average annual rate of 27.53 percent and 30.49 percent, respectively.

Similarly, meta-learning was employed for sales forecasting was explored in [17]. For the sales forecast, the meta-learning architecture used deep convolutional neural networks. The neural networks automatically learn a feature representation from raw sales time series data, then connect those features to a collection of weights that are used to integrate a pool of base-forecasting methods. The experiments using IRI weekly data show that the proposed meta-learner outperforms a number of state-of-the-art benchmarks in forecasting, though the accuracy gains over some more sophisticated meta ensemble benchmarks are modest, and the learned features are unintelligible. However, authors in [18] observed that the effects of marketing strategies over sales can change anytime resulting in structural change problem. As a result, more accurate approaches for forecasting retailer inventory prices that account for structural change have been suggested. The proposed approach outperforms traditional forecasting approaches that neglect the probability of such improvements, according to data from a well-known US retailer. A sales forecasting model for computer products manufacturers was proposed in [19]. In the meanwhile, sales forecasting's success is heavily influenced by the complex relationships between the major competitors. The research presents a novel framework to overcome difficulties such as: (1) seasonal and cyclic patterns in computer product demand are recorded, (2) sales forecasting consider product uncertainty changes, and (3) the complex relationships between the manufacturers and computer products are considered to evoke managerial insights. Experiments demonstrate that the proposed framework accomplishes the above objectives and has the ability to be applied to other manufacturing components. The application of deep neural networks for sales forecasting in the fashion industry was carried out in [20]. The models were created using a broad range of factors, in-

cluding product physical characteristics and domain experts' opinions. Furthermore, this research contrasts the revenue forecasts obtained using a range of shallow techniques, such as Decision Trees, Random Forest, Support Vector Regression, Artificial Neural Networks, and Linear Regression, with those obtained using a collection of deep learning techniques. The deep learning model was found to have strong results in predicting sales in the fashion retail industry, but it does not do substantially better than some of the shallow strategies, such as Random Forest, for some of the measurement metrics considered. However, this study explore how XGBoost can be used to forecast sales. Walmart stores sales data that was collected from 45 stores were sued to train and test the model.

3 Methodology

The process flow involved in developing the model is shown in Fig. 1. It is necessary that each technique goes through the build up to generate accurate results. There are about six steps involved and each one plays a crucial role to build up the proposed model. The pre-processing part are the first three part of the steps before the model was built. The single model was developed using XGBoost algorithm. The performance evaluation gives the performance of the model which was obtained using the testing set the performance metrics.

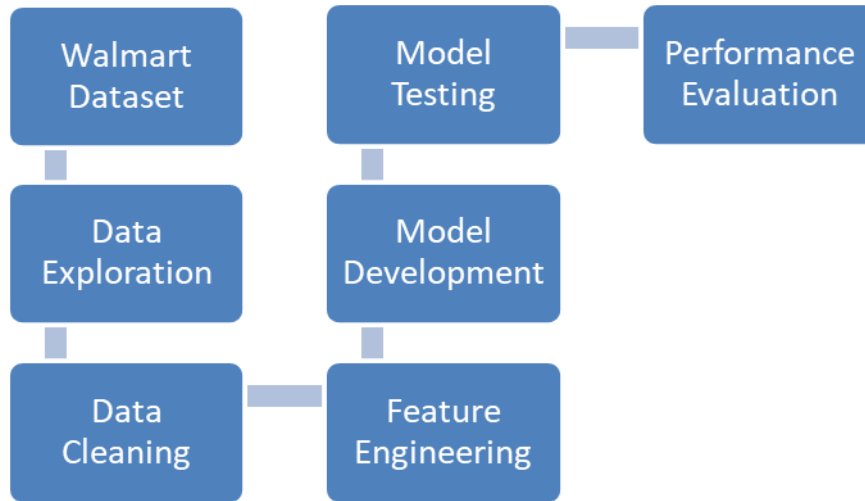


Fig. 1. Flow Diagram of Processed System

3.1 Data Collection

The data used in this study was collected from 45 Walmart stores in the United States. They are the record of sales made in their 45 branches between from 2010-02-05 to 2012-11-01. Each store is made up of several departments and our task is to predict the department-wide sales for each store. Walmart did not provide much information about what is the type of the store or the units of the store size. The datasets consist of four different files namely.

- stores.csv: This file includes anonymized data about the 45 stores, including store type and capacity.

- **train.csv:** This is the training data from 2010-02-05 to 2012-11. The file has the following fields: Store (the store number), Dept (the department number), Date of the week, Weekly_Sales (sales for the given department in the given store), IsHoliday (whether the week is a special holiday week or not).
- **test.csv:** This file is similar to train.csv, with the exception of the weekly sales area, which must be expected. Our goal is to forecast revenue by store, department, and date.
- **features.csv:** For the dates specified, this file provides additional information about the shop, department, and regional operation. It has the following sections: a place to keep things (the store number), the week's day Thermometer (average temperature in the region), Fuel Cost (cost of fuel in the region), Markdown1-5 (anonymized data linked to Walmart's promotional markdowns); Markdown data is only available after November 2011 and is not always available for all stores. IsHoliday, CPI, unemployment (the unemployment rate) (whether the week is a special holiday week or not). N/A is used to indicate any missed values.

3.2 Data Exploration

Given the dataset, key features of the dataset that will be used for the prediction needs to be extracted. These are features about the stores: store unique ID, store type, store size, temperature of the region during the week, fuel price in the region during a particular week, unemployment rate during that week in the region of the store, date, Weekly_sales, CPI, isHoliday and Markdowns. These extracted features will be used for the training and testing of the model.

3.3 Data Cleaning

In this phase of the study, the null values present in the dataset were removed. While exploring the dataset, it was observed that the markdown 1-5 contains some null values. To know if the features will be useful in the model development, a threshold of 60% was set as an acceptable percentage for null values. The number of missing values from the markdown columns and their percentage over the total of registers were counted. However, it was observed that 60% of the data from all markdown columns are missing, therefore the columns were discarded while other columns whose null values does not exceed the set threshold were converted to 0. The IsHoliday column was also transformed to numeric values 0 for False and 1 for True.

3.4 Feature Engineering

Feature engineering refers to a process of selecting and transforming features when a predictive model is created by machine learning or statistical modelling. Feature engineering technique used in this study are:

- **One-Hot Encoding:** One of the most popular encoding methods in machine learning is one-hot encoding. This method assigns 0 or 1 to several flag columns based on the values in a column. The relationship between the grouped and encoded columns is represented by these binary values. This technique was used to split the type column and create separate columns for each of the types and assign 0 or 1 to each column depending on the type column this was also used to convert the types column from char to integer and give the columns some form of hierarchy
- **Extracting Date:** Even though date columns provide valuable information on the model objective, the information is disregarded as an input. This may be because dates can be found in numerous formats that make algorithms difficult to understand, even in formats such as "01-01-2017". If the date columns are not re-arranged, it will be difficult for a machine learning algorithm to build an ordinal relationship between the values. Therefore, the different sections of the dates were separated into different columns: year, month, and day. With this, the date information becomes easy for the machine learning algorithm to manipulate.

3.5 Model Building using XGBoost Algorithm

The next task is model building using the XGBoost algorithm. This was implemented using the XGBoost library. The library contains all the tools needed to build different XGBoost model. It is an open-source implementation of the gradient boosting algorithm that is quite effective. As such, XGBoost is a Python library, an open-source project, and an algorithm. It is designed to be both computationally efficient (i.e., fast to execute) and highly effective, perhaps even more so than other open-source implementations. Execution speed and model consistency are the two key factors to use XGBoost. On classification and regression predictive modelling issues, the algorithm's main goal is to maximize the objective function's value. Rather than using feature vectors to measure the similarities between forecasting and history days, gradient boosting builds boosted trees to obtain feature scores intelligently, showing the value of each feature to the training model. The higher a feature's score becomes, the more it is used to make crucial decisions with boosted trees. The algorithm divides the value into three categories: "benefit," "frequency," and "cover". The key criterion for determining the value of a function in the tree branches is gain. Frequency is a simplified form of gain. The dataset is then split into two sets: a training set and a testing set. The training set is used to train the model during the development process, while the testing set is used to assess the model's results. The dataset was split into two parts in this report: 70% for the training set and 30% for the research set.

XGBoost Algorithm

XGBoost is a modular tree boosting machine learning framework. The software can be downloaded as an open-source kit. The technology has had a big influence and is well-known in a variety of deep learning and data mining problems. The scalability of XGBoost in all situations is the most important explanation for its success. On a single processor, the technology is ten times faster than current common implementations, and it scales to billions of examples in distributed or memory-limited environments. XGBoost's scalability is attributed to many significant machine and algorithmic optimizations, including a novel tree learning algorithm for sparse data handling and a technically justified weighted quantile sketch technique for estimated tree learning instance weight handling. Parallel and distributed processing speed up research, allowing for more rapid model discovery. More specifically, the model takes advantage of out-of-core computing to enable data scientists to process a hundred million examples on a single computer. Finally, after integrating these methods to create an end-to-end method, it can expand to even more data by using the fewest cluster resources possible. We used the XGBRegressor from the xgboost open-source kit available in Python. We set our parameter to the following after tuning the XGBoost model several times as proposed in (Truong et al., 2020):

- Set the learning rate parameter to 0.1.
- Set the number of estimators to 200.
- Set regularization parameters: reg lambda = 0.45, reg alpha = 0, gamma = 0.5, min child weight = 2, subsample = 1, colsample bytree = 0.8.

3.6 Model Evaluation

The performance of the proposed prediction model was carried out using the following metrics:

- The Root Mean Square Error (RMSE) is the residuals' standard deviation (prediction errors). Residuals are an indicator of how far data points are out from the regression line; RMSE is a measure of how these residuals are spaced out. It calculates the degree to which the data is concentrated in the best fit axis. It was determined using equation (1):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (S_i - O_i)^2} \quad (1)$$

- Mean absolute error (MAE): tests how reliable the estimated values are relative to the actual ones. It was calculated using equation (2):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y(i) - \hat{y}(i)| \quad (2)$$

- Coefficient of Determination or R^2 uses two mean squared error calculations. The former is the mean squared error of the current and expected values, and the latter is the mean squared error of each real value and the sum of observations. It was determined using equation (3):

$$R2 = \frac{\sum_{i=1}^m (y(i) - \hat{y}(i))^2}{\sum_{i=1}^m (y(i) - \bar{y})^2} \quad (3)$$

4 RESULTS AND DISCUSSION

The results obtained from the model implementation is discussed in this section.

4.1 Results of the Data Exploration

Data exploration was carried out to understand the features present in the dataset and their purpose. The distribution of the train dataset is shown in Fig. 2. This shows the distribution of the data in each column such as the mean, standard deviation, the count and other values. The next phase was to check the number of null values in the train data. However, there was no null value in the train data

	Store	Dept	Weekly_Sales
count	421570.000000	421570.000000	421570.000000
mean	22.200546	44.260317	15981.258123
std	12.785297	30.492054	22711.183519
min	1.000000	1.000000	-4988.940000
25%	11.000000	18.000000	2079.650000
50%	22.000000	37.000000	7612.030000
75%	33.000000	74.000000	20205.852500
max	45.000000	99.000000	693099.360000

Fig. 2. Distribution of the Train Dataset

Afterwards, the distribution of the features dataset was also explored as shown in Fig. 3 and the presence of null values were examined in the columns.

	Store	Temperature	Fuel_Price	MarkDown1	MarkDown2	MarkDown3	MarkDown4	MarkDown5	CPI	Unemployment
count	8190.000000	8190.000000	8190.000000	4032.000000	2921.000000	3613.000000	3464.000000	4050.000000	7605.000000	7605.000000
mean	23.000000	59.356198	3.405992	7032.371786	3384.176594	1760.100180	3292.935886	4132.216422	172.460809	7.8268
std	12.987966	18.678607	0.431337	9262.747448	8793.583016	11276.462208	6792.329861	13086.690278	39.738346	1.8772
min	1.000000	-7.290000	2.472000	-2781.450000	-265.760000	-179.260000	0.220000	-185.170000	126.064000	3.6840
25%	12.000000	45.902500	3.041000	1577.532500	68.880000	6.600000	304.687500	1440.827500	132.364839	6.6340
50%	23.000000	60.710000	3.513000	4743.580000	364.570000	36.260000	1176.425000	2727.135000	182.764003	7.8060
75%	34.000000	73.880000	3.743000	8923.310000	2153.350000	163.150000	3310.007500	4832.555000	213.932412	8.5670
max	45.000000	101.950000	4.468000	103184.980000	104519.540000	149483.310000	67474.850000	771448.100000	228.976456	14.3130

Fig. 3. Distribution of the Feature Dataset

However, there are null values in the Markdown1 – Markdown5 columns as well as the CPI and unemployment columns. The remaining datasets were also explored and the null values were removed. The next stage of the exploration is the combining the datasets together to get one dataset that is going to be cleaned and be used to train the model that will be built in the report. First, we will merge the train dataset and feature dataset on the columns that are common in both the datasets and that is the Store, Date and IsHoliday columns and it going be an inner join after joining them, then we are going to join the Store dataset to the new dataset that is the combination of the train dataset and the feature dataset, they will be combined on the Store column and also an inner join after joining all the dataset an new dataset is created that contains all the columns from the train, feature and the store datasets. After that count plot of the number of sales per store was plotted as shown in Fig. 4. This was also done to get a better knowledge of the relationship among related features in the dataset.

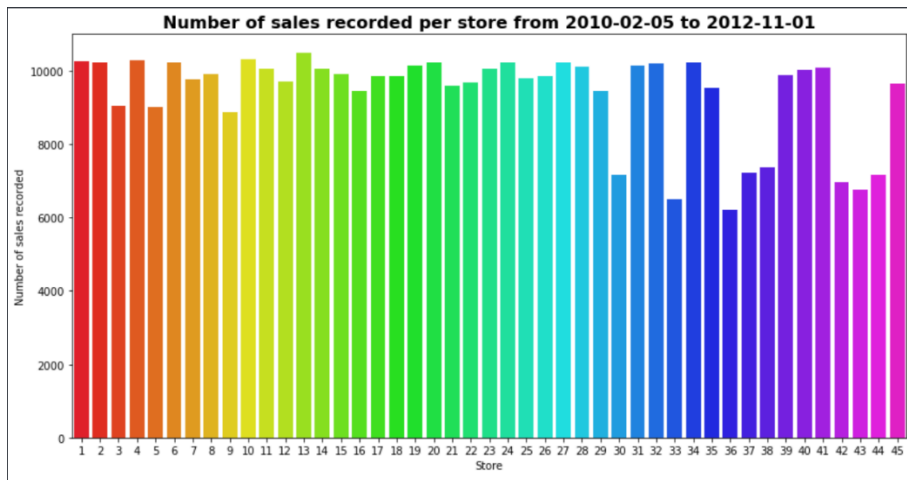


Fig. 4. Count Plot of Number of sales per Store

The histogram of the input dataset was also generated so as to examine the frequency of sales during the time of the data collection. Fig. 5 shows the total number of sales generated by each of the 45 stores during the time of the collection of the dataset. It was observed that more stores have between 8000 and 10000 sales during this period.

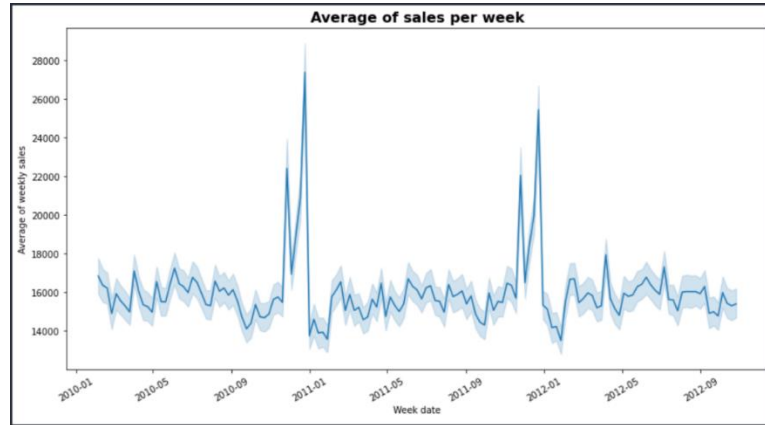


Fig. 5. Average Sales Per Week

This line plot shows the average sales of all the store is per week for the years the dataset was also explored as shown in Fig. 6. It was observed that in the latter part of 2010 the stores made more sales than any other year followed by the latter part of 2011.

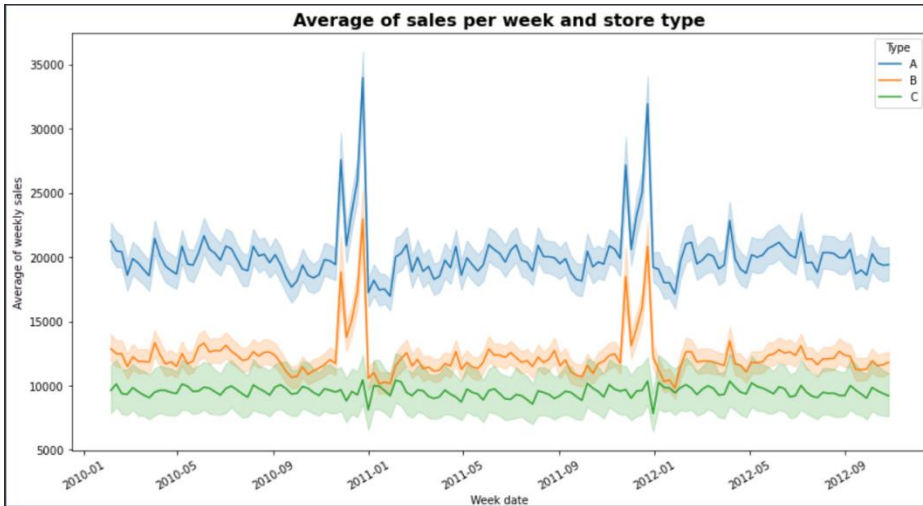


Fig. 6. Line Plot of Average Sales Per Week and Store Types

This line plot shows the average sales for each type of the stores which are type A, B, C. type a made the most sales and type c made the least sales for the duration of the dataset.

4.2 Performance Evaluation of the Proposed Model

In this stage after training and testing the model, Mean Absolute Error (MAE), R2, Mean Square Error (MSE) and Root Mean Square Error (RMSE) were computed as shown in Fig.7. The results obtained revealed that XGBoost achieved an appreciable prediction accuracy.

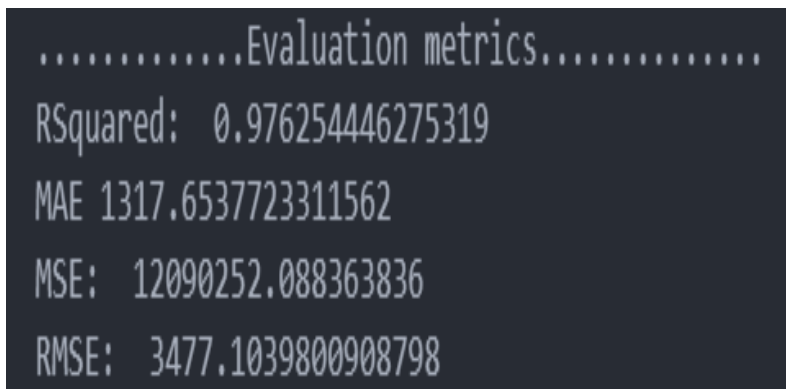


Fig. 7. Evaluation of XGBoost Model

5 CONCLUSION

In supply chain, sales forecasting is a big concern and a critical role. This significance stems from the fact that it has an effect at all levels of the organization. A company's inaccuracy or lack of forecasting may result in inadequate inventory and material flow control, revenue losses or surplus inventory, and consumer frustration. When forecasting issues are combined with other supply chain flaws like the bullwhip effect, the effects may be disastrous. While research and practice favor historical sales forecasting techniques, causal methods have been shown to be more accurate and precise. For reliable revenue forecasting, extensive analysis is being conducted at the enterprise stage. Since a company's profit is directly proportional to its ability to forecast profits accurately, Big Marts are looking for a more reliable prediction algorithm to ensure that the company does not lose money. This study explore how Extreme Gradient Boosting (XGBoost) can be used to forecast sales. Walmart stores sales data that was collected from 45 stores were sued to train and test the model. Results obtained from the mean absolute error, R2, mean square error and root mean square error revealed that XGBoost machine learning technique can be effectively used to predict sales and this could guide sales managers and other practitioners in making decisions about how to fix prices of goods.

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